CH 4 - Multivariate Discrete Random Variable

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4.1 Basic Terminology and Techniques

Ofter we are interested in multiple phenomenal that vary alongside each other.

Examples:

- Treatment/dose and disease status
- Stock price or credit rating of 2 companies
- Pollutant level and respiratory illnesses

> 📘 **Definition — Joint Probability Mass Function**
>
> The **joint probability mass function** of two random variable X and Y is
>
> $$f(x,y) = P(X = x, Y = y)$$
>
> for (x, y) in joint rnage of X, Y respectively.
>
> It is represented usually by a table, occasionally a function, never a histogram.

Properties for JPMF:

1. $0 \leq f(x,y) \leq 1$
2. $\sum_x \sum_y f(x,y) = 1$

We can find any probability involving x and y by summing up the $f(x,y)$ over all pairs that satisfy

Example:

Flip a fair coin 3 times and let $X = \#$ of heads obtained, $Y = \begin{cases} 1 \text{ if first flip is T} \\ 0 \text{ else} \end{cases}$. Find $f(x,y)$

	0	1	2	3
0	0	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$
1	$\frac{1}{8}$	$\frac{2}{8}$	$\frac{1}{8}$	0

$P(2 < X + Y) = f(3,0) + f(2,1) + f(3,1) = \frac{2}{8}$

$P(X - Y = 1) = f(1,0) + f(2,1) = \frac{2}{8}$

> 📘 **Definition — Marginal PMF**
>
> The marginal PMF of X is
>
> $$f_X(x) = P(X = x) = \sum_y f(x,y)$$
>
> $$f_Y(y) = P(Y = y) = \sum_x f(x,y)$$

In general, the JPMF of $X_1, X_2, ..., X_n$ is $f(x_1, x_2, ..., x_n) = P(X_1 = x_1, X_2 = x_2, ..., X_n = x_n)$

Marginal PMF of X_1 is $f_1(x_1) = \sum_{X_2} ... \sum_{X_n} f(x_1, x_2, ..., x_n)$.

The JPMF of X_1, X_2 is $f_{1,2}(x_1, x_2) = \sum_{X_3} ... \sum_{X_n} f(x_1, x_2, ..., x_n)$

📘 **Definition — Independence Random Variable**

We say two random variables X, Y are independent if and only if

$$f(x, y) = P(X = x, Y = y) = P(X = x)P(Y = y) = f_X(x) + f_Y(y)$$